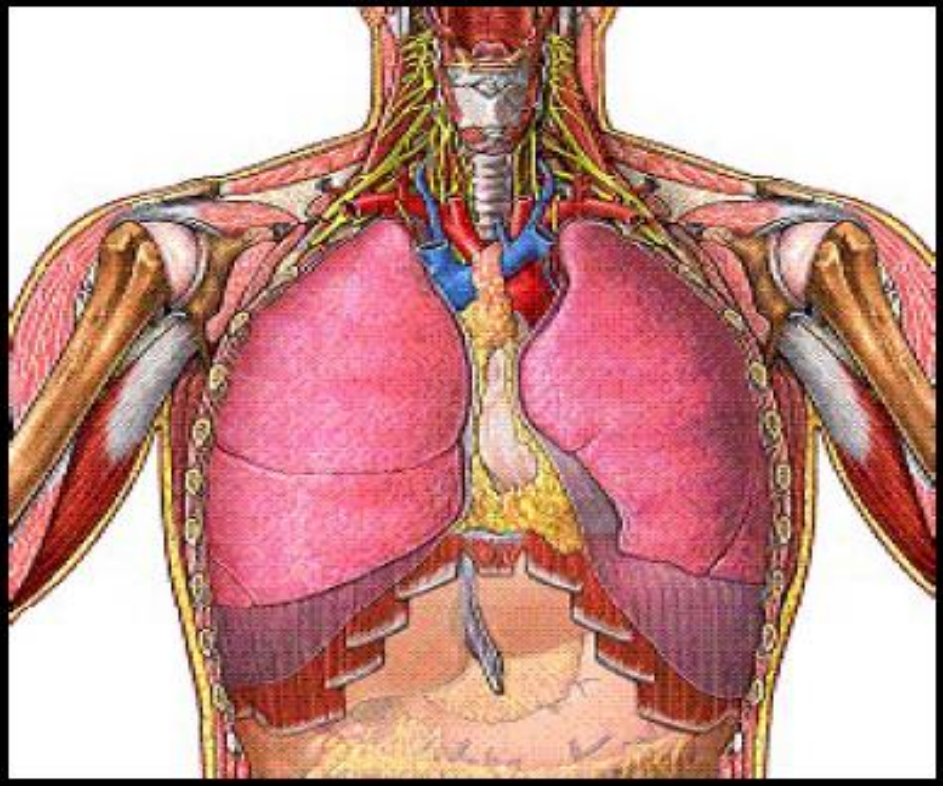
THE PARTS AND FUNCTIONS OF







HUMAN RESPIRATORY SYTEM

Main functions of Respiratory System

- Providing an extensive surface area for gas exchange between air and circulating blood.
- Moving air to and from the exchange surfaces of the lungs along the respiratory passageways.

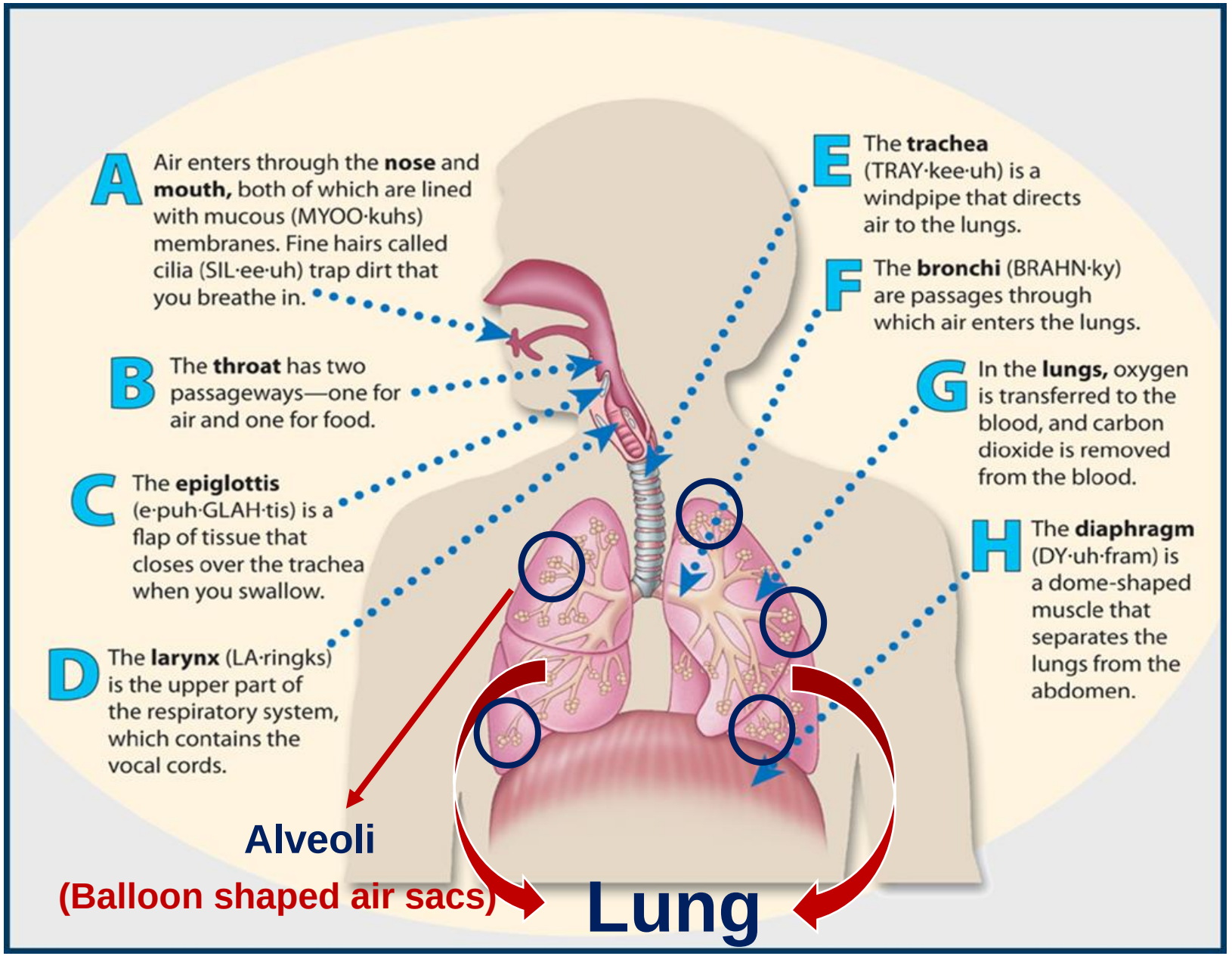
1. The respiratory system provides oxygen for metabolism in the tissues.

2. The respiratory system removes carbon dioxide, the waste product of metabolism.

- Protecting respiratory surfaces from dehydration, temperature changes, or other environmental variations, and defending the respiratory system and other tissues from invasion by pathogens.
- Producing sounds involved in speaking, singing, and other forms of communication.
- Facilitating the detection of olfactory stimuli by olfactory receptors in the superior portions of the nasal cavity.

Organs of Respiratory System

- Nose
- Pharynx
- Larynx
- Trachea
- Bronchi
- Lungs – alveoli



Organs in the Respiratory System

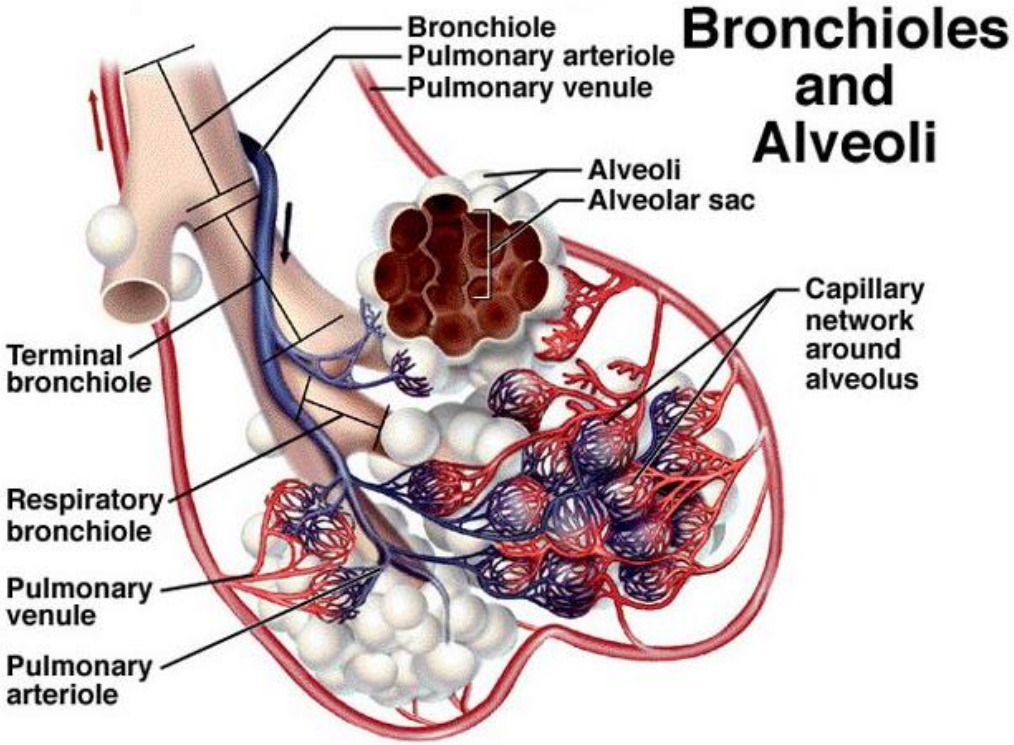
STRUCTURE	FUNCTION
Nose / nasal cavity	Warms, moistens, & filters air as it is inhaled
Pharynx (throat)	Passageway for air, leads to trachea
Larynx	The voice box, where vocal cords are located
Trachea (windpipe)	Trachea is lined with fine hairs called *cilia* which filter air before it reaches the lungs
Bronchi	Two branches at the end of the trachea, each lead to a lung
Lungs Bronchioles	A network of smaller branches leading from the bronchi into the lung tissue & ultimately to air sacs
Alveoli	The functional respiratory units in the lung where gases are exchanged
Diaphragm	A dome-shaped muscle that separates the lungs from the abdomen

Functional division of respiratory system

Two parts:

1) **Conductive zone**: main function is to filter, warm, and moisten air and conduct it into the lungs. These include nose, nasal cavity, pharynx, larynx, trachea, bronchi, bronchioles.

2) **Respiratory zone**: main sites for gas exchange between air and blood. These include bronchioles, alveolar ducts, alveolar sacs and alveoli.



Respiration process: Breathing and cellular respiration

What is breathing?

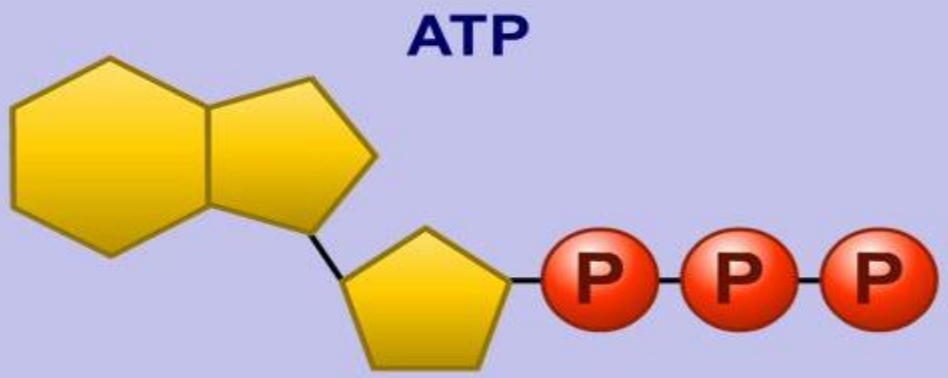
- It is a physical process by which organisms obtain oxygen from the air and release carbon dioxide.

What is cellular respiration?

Respiration is the process by which organisms extract the energy stored in complex molecules and use it to generate **adenosine triphosphate (ATP)**.

In this way they obtain energy to fuel their metabolic pathways.

ATP provides the immediate source of energy for biological processes such as active transport, movement and metabolism.



Respiration process and its four events

Breathing	Gaseous exchange at lungs level	Gas transport by blood	Gaseous exchange at tissue level	Cellular Respiration
Air movement into and out of lungs	Exchange of gases between alveoli and blood	Transport of oxygen from blood capillaries of alveoli to body cells and return of carbon dioxide	Exchanging of oxygen from blood into the cells and carbon dioxide from cells into the blood	Using oxygen in cell processes to produce carbon dioxide and water, releasing energy to be used for life processes

Pulmonary ventilation

External respiration or pulmonary exchange of gases

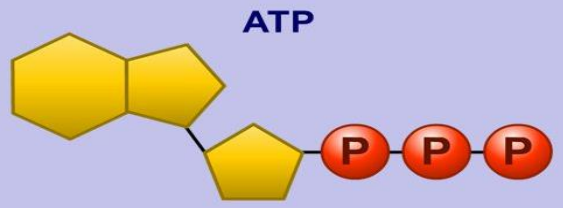
Respiratory gas transport

Internal respiration or Systemic exchange of gases

Respiration is the process by which organisms extract the energy stored in complex molecules and use it to generate **adenosine triphosphate (ATP)**.

In this way they obtain energy to fuel their metabolic pathways.

ATP provides the immediate source of energy for biological processes such as active transport, movement and metabolism.



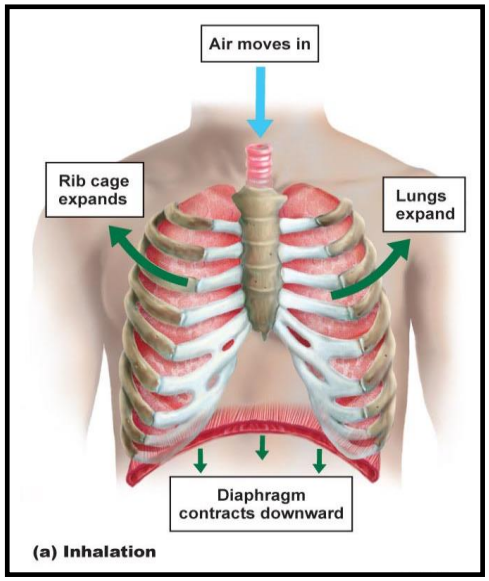
Pulmonary Ventilation/Breathing

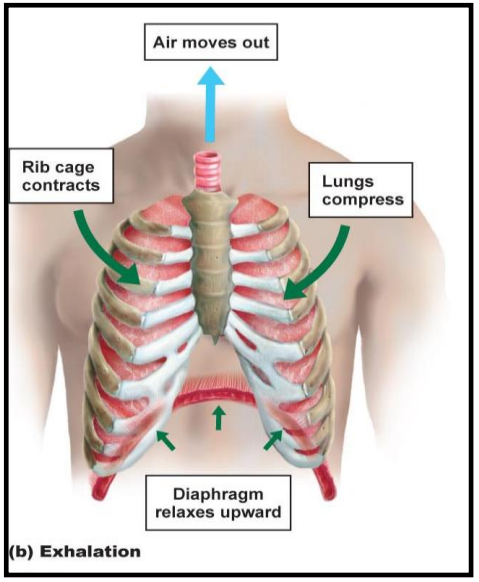
- **Pulmonary ventilation** is a mechanical process depends on volume changes in the thoracic cavity. Volume changes lead to pressure changes, which leads to the flow of gases to equalize pressure.
- Two phases of pulmonary ventilation/breathing
  - **Inhalation (Inspiration), and**
  - **Exhalation (Expiration)**

Content	Inhaled Air
Oxygen	21 %
Carbon dioxide	0.04 %

Content	Exhaled Air
Oxygen	16 %
Carbon dioxide	4 %

Inspiration:During inspiration (inhalation), the diaphragm and intercostal muscles contract due to which there is expansion of rib cage and lungs and increase in thoracic cavity. External air is pulled into the lungs due to increase in the volume of thoracic cavity.





Exhalation: During exhalation, intercostal muscles relax. The diaphragm moves upwards due to which rib cage contracts and lungs are compressed. As a result, air is pushed out of the lungs.

Inhalation Process



Air

Breathing in

Enters via nose or mouth

Travels down to

Pharynx and Larynx

Primary filtration done

Enters into

The Trachea

Enters into

Left and Right Bronchi

Further filtration done

Passes through

Bronchioles

Finally enters into

The Alveoli

Absorption of oxygen from the air

Diffusion of oxygen into

The Blood Capillaries

External respiration and internal respiration

❖ **External respiration / Pulmonary exchange of gases** is the exchange of gases between oxygen in the air and blood capillaries in the alveoli. Diffusion of gases **between lungs-alveoli and blood.**

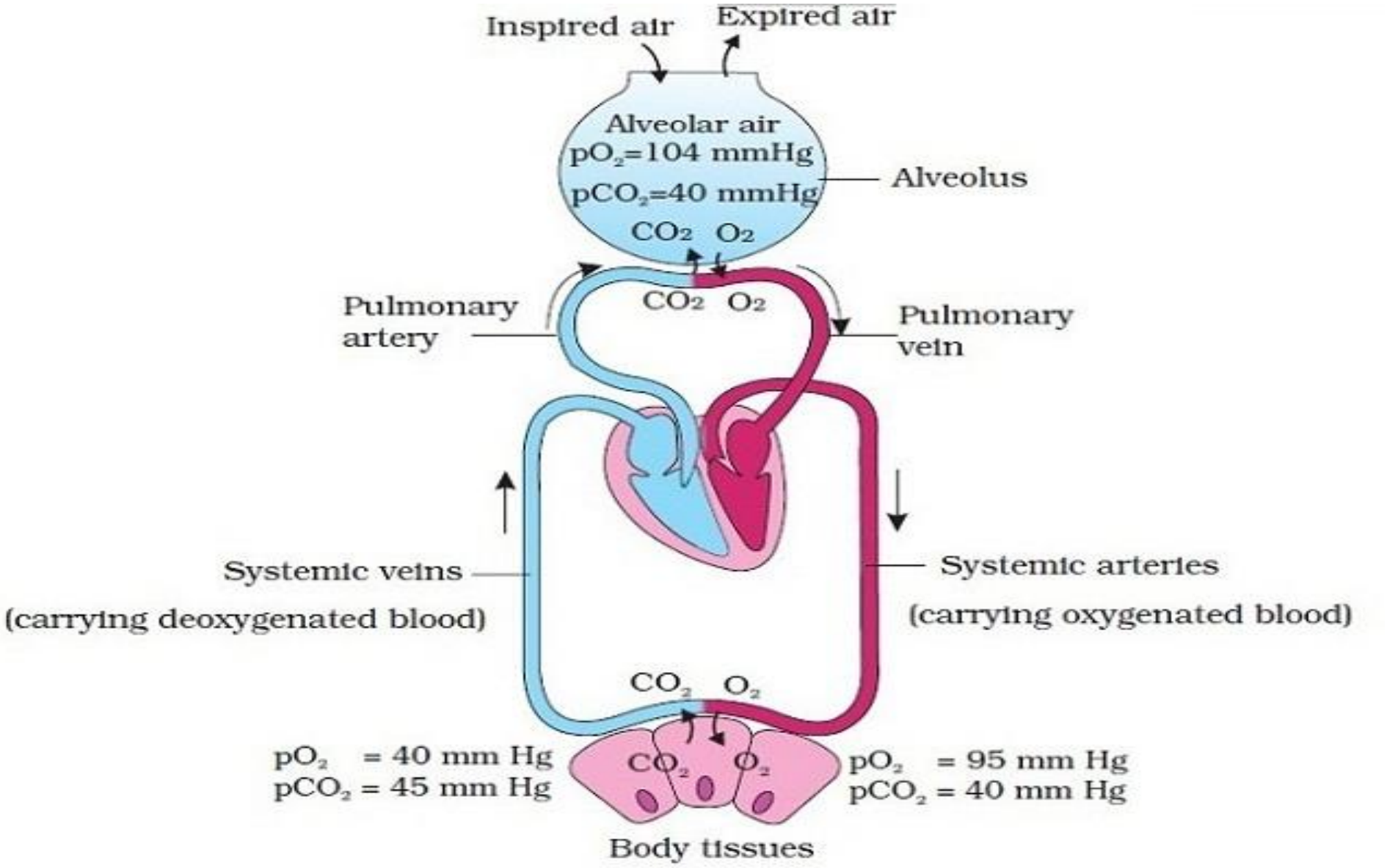
- Oxygen diffuses from alveoli into blood.
- Carbon dioxide simultaneously diffuses in opposite direction.

Blood has low PO_2 (40 mm Hg) while PO_2 in alveoli air is 104 mm Hg.

Blood has high CO_2 (45 mm Hg) compared to alveoli air (40 mm Hg).

❖ **Internal respiration/Systemic exchange of gases** is the exchange of gases between blood and body tissues. Cells use oxygen constantly for cellular respiration so partial pressure in tissue is low. Tissue produce large quantities of CO_2 as biochemical waste products so partial pressure is high.

- **Pulmonary exchange of gases (between lungs and blood) and Systemic exchange of gases (between blood and body tissues)**



Transport of respiratory gases

1) Oxygen transport from lungs to body tissues

2) Carbon-dioxide transport from body tissues to lungs

RESPIRATORY GASES; How are they transported in the blood???



OXYGEN, O_2

The only way for Oxygen to be carried around the body is by the HEAMOGLOBIN, in red blood cells.







CARBONDIOXIDE, CO_2

Carbondioxide is transported by the blood in three ways;

A)As sodium hydrogencarbonate in plasma

B)As carbomino compounds in red cells

C) In simple solution, dissolved in the plasma





Transport of respiratory gases in the blood

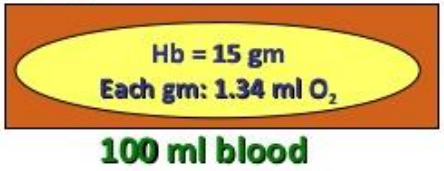
Oxygen transport

- **O_2 + Fe^{2+}-Haemoglobin (Hb) → HbO_2 (oxyhemoglobin)---98.5%**
- **Oxyhemoglobin** carries 1-4 molecules of oxygen depending on its degree of saturation.
- **Oxyhemoglobin** does not dissociate in the blood vessel, until it reaches a tissue with low Po_2 (partial pressure of oxygen).
- **In dissolved state**, only 3% oxygen is transported to the tissues.

O_2 CARRYING CAPACITY OF THE BLOOD

- *It is the maximum amount of O_2 that can be carried by Hb.*
- *Each gram Hb, when fully saturated with O_2, can carry 1.34 ml O_2.*
- *As Hb content = 15 gm / 100 ml blood.*

So, O_2 carrying capacity = 1.34 x 15



= 20.1 ml O_2/100 ml blood.

Carbon-dioxide transport from tissues to lungs

1. In dissolved state in plasma-5-7% is transported in dissolved state. Upon reaching the lungs, it diffuses into alveolar air and exhaled.

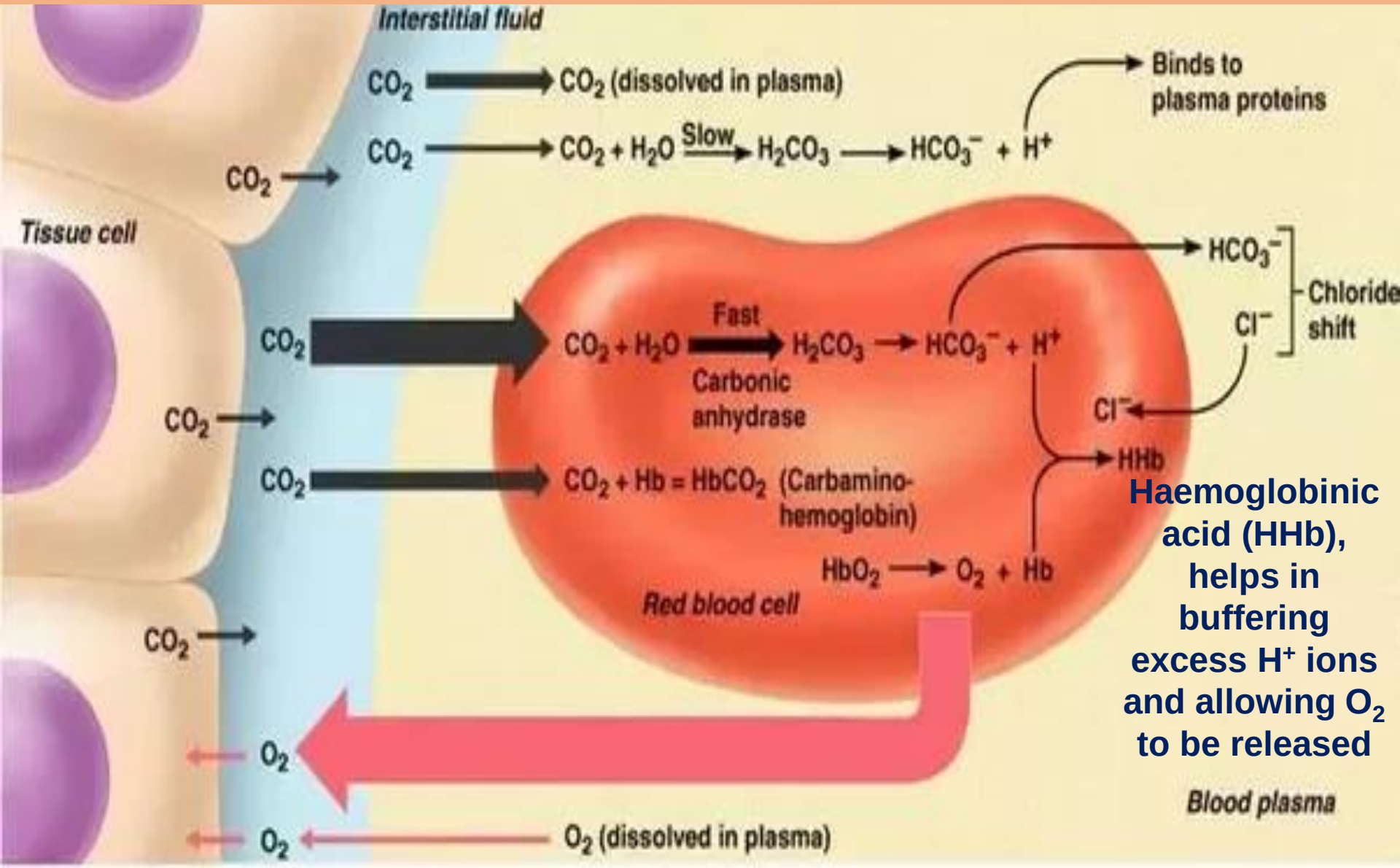
2. Bicarbonate ions: 70% CO_2 is transported in the form of bicarbonate ions.

- CO_2 (from tissues) + H_2O (water in erythrocytes) $\Leftrightarrow$ H_2CO_3 (Carbonic acid) **reaction is catalyzed by enzyme Carbonic anhydrase which is present in erythrocytes.**
- **$H_2CO_3 \Leftrightarrow H^+ + HCO_3^-$**
- Carbonic acid that is formed immediately dissociates into Bicarbonate ions (HCO_3^-). Bicarbonate ions diffused into the plasma and reacts with sodium ions to form **sodium bicarbonate ($NaHCO_3^-$).** In exchange of HCO_3^-, chloride ions move from plasma into the RBCs. This exchange of negative ions, which maintains the electrical balance between blood plasma and RBC cytosol, is known as chloride shift**.**

3. Carbamino-haemoglobin- 23% of CO_2 reacts with amine group of globin part of haemoglobin (Hb) forming carbamino-haemoglobin ($HbCO_2$)

Gas exchange at tissues level

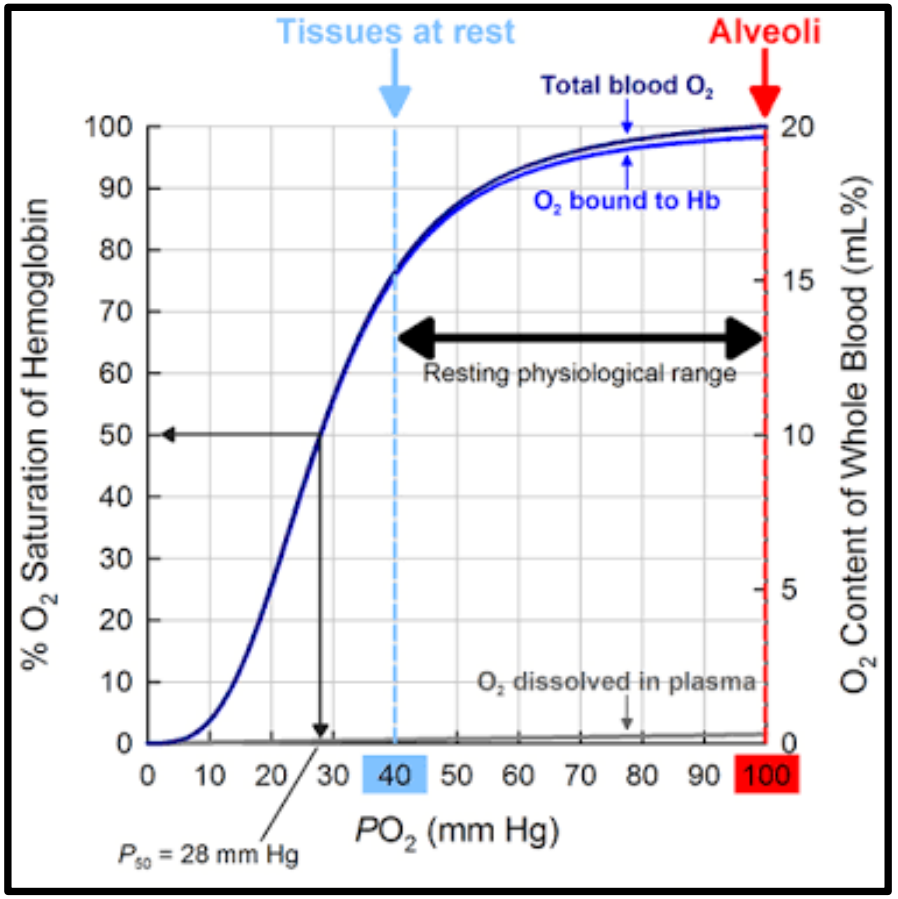
(Oxygen release and carbon dioxide pick up)



Haemoglobinic acid (HHb), helps in buffering excess H^+ ions and allowing O_2 to be released

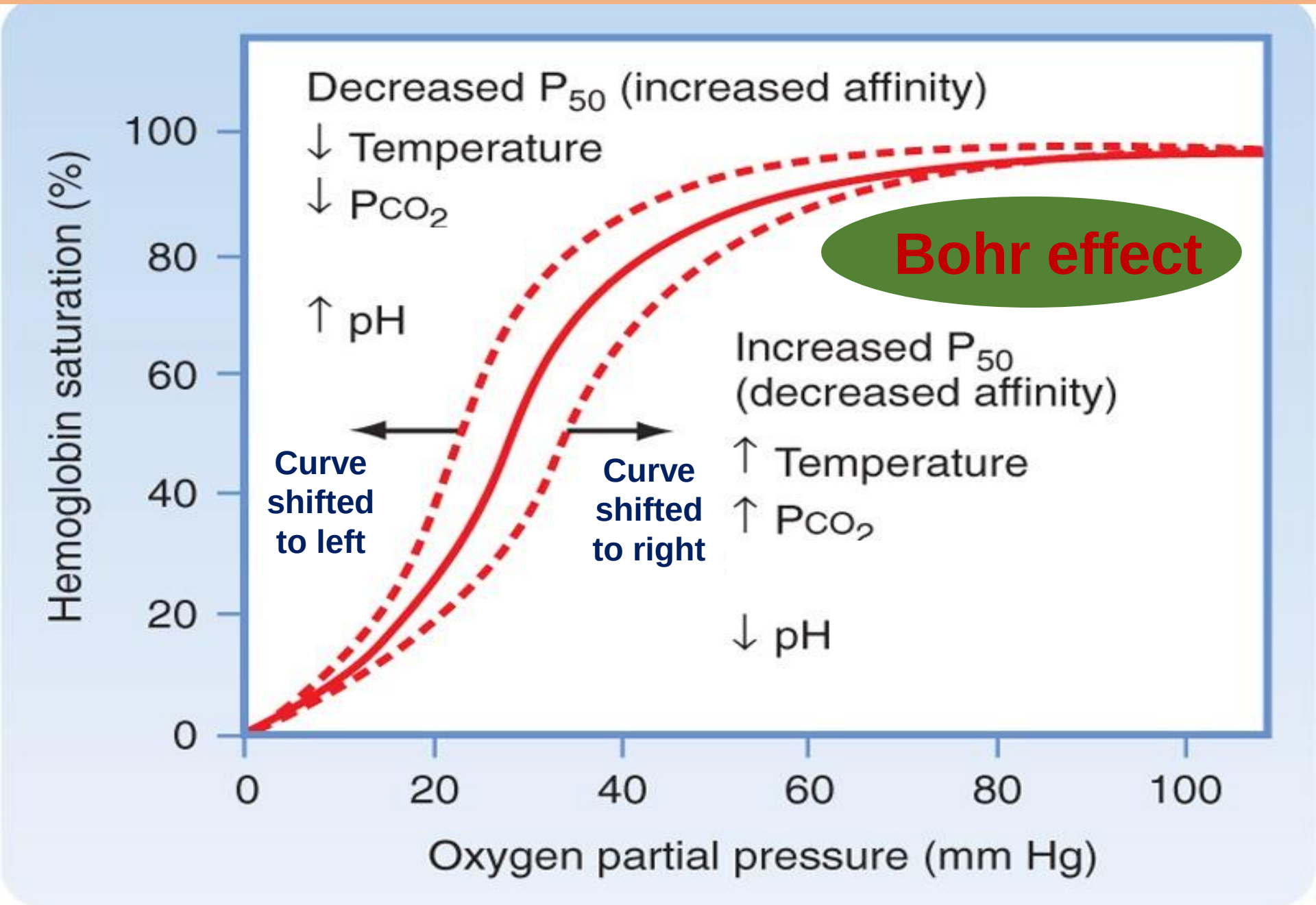
Oxygen-haemoglobin dissociation curve (ODC) showing the relationship between hemoglobin saturation and Po_2 at normal body temperature

- **When** Po_2 **is** high, the more O_2 combines with hemoglobin and is almost 100% saturation.
- **When** Po_2 is low, hemoglobin is only partially saturated.
- **In pulmonary capillaries**, where Po_2 is high, a lot of O_2 binds to hemoglobin. In tissue capillaries, where the Po_2 is lower (40 mm Hg), hemoglobin does not hold as much oxygen and dissolved Oxygen is unloaded via diffusion into tissue cells.
- **Hemoglobin is** still 75% saturated with O_2 at a Po_2 of 40 mm Hg, the average Po_2 of tissue cells in a person at rest. That means only 25% of the available O_2 unloads from hemoglobin and is used by tissue cells under resting conditions.



Haemoglobin-oxygen binding affinity

Factors that causes shifts in the ODC - left and right shift



Respiratory system sample questions

1. **Which of the following is NOT the function of the respiratory system?**
a) Regulate blood pH
b) Helps in gaseous exchange
c) Sound production
d) Contains receptors for the sense of smell

2. **Which of this statement is TRUE for pulmonary respiration?**
a) Exchange of gases between alveoli of lungs and the blood
b) Exchange of gases between blood and tissue cells
c) Breathing between the atmosphere and the alveoli of the lungs
d) Production of ATP

3. **Oxygen and hemoglobin bind in a reversible manner to form_____**
a) Carboxyhemoglobin
b) Oxyhemoglobin
c) Methoglobin
d) Haemoglobinic acid

4. **How many oxygen molecules bound to hemoglobin to give 50% saturation?**
a) 6
b) 4
c) 2
d) 7

5. **Dissociation curve shifts to the right when**

a) CO_2 concentration decreases
b) CO_2 concentration increases
c) O_2 concentration decrease
d) CL^- concentration increase